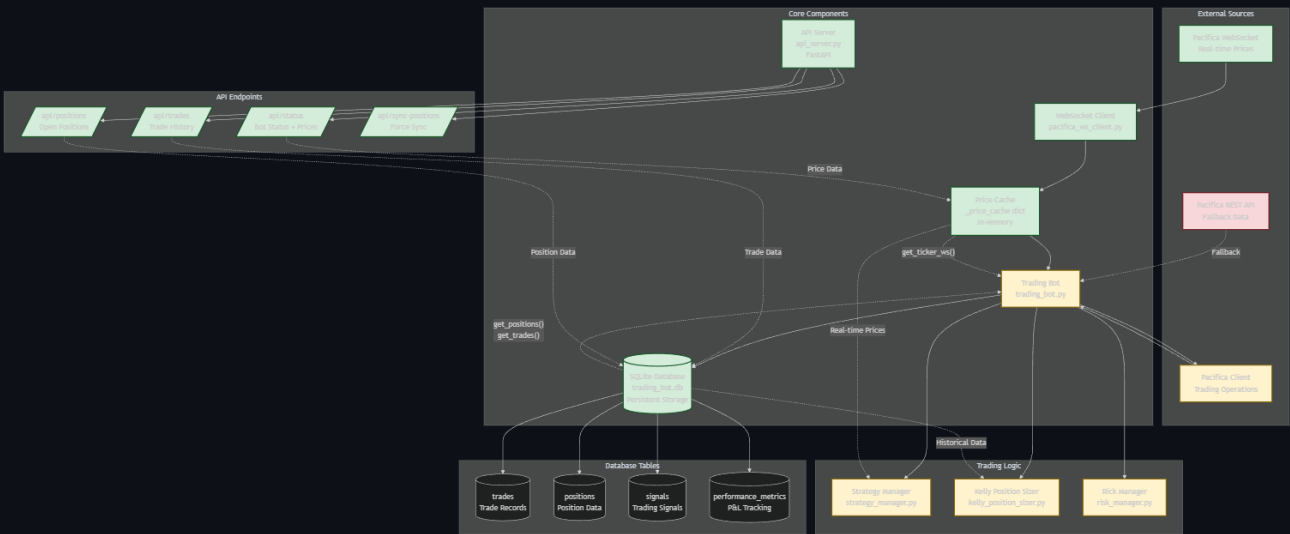


Trading Bot Workflow Diagram



Current State Analysis

✅ Working Components

- **WebSocket Client:** Successfully connects and parses price updates
- **Price Cache:** Stores real-time prices in memory for fast access
- **Database:** SQLite setup with all required tables (trades, positions, signals, metrics)
- **API Server:** FastAPI endpoints serving status, trades, positions
- **Database Tables:** All 15 tables created and functional

⚠️ Partially Working

- **Trading Bot:** Core logic exists but needs integration testing
- **Pacifica Client:** Trading operations interface (needs verification)
- **Strategy Manager:** Signal generation logic (needs testing)
- **Kelly Position Sizer:** Risk-adjusted sizing (depends on trade history)
- **Risk Manager:** Circuit breaker and position limits (needs validation)

❌ Needs Fixes

- **REST API Fallback:** Should be backup only, not primary data source
- **WebSocket Price Integration:** Ensure `get_ticker_ws` prioritizes cache over REST
- **Strategy Execution:** Verify signals trigger actual trades
- **Position Sync:** Ensure database positions match Pacifica state

Critical Integration Points

1. Price Data Flow

WebSocket → Price Cache → Trading Bot → Strategies

Status: WebSocket parsing fixed, cache working, needs verification that strategies use cache.

2. Trade Execution Flow

Strategy Signal → Position Sizing → Risk Check → Client Order → Database Save

Status: Components exist but end-to-end flow needs testing.

3. Data Persistence Flow

Live Positions → Database → API Endpoints → Monitoring

Status: Database saves working, API serving data, needs sync verification.

Required Fixes for Full Integration

- 1. **Verify Price Priority:** Ensure `get_ticker_ws` returns WebSocket cache data, not REST fallback
- 2. **Test Strategy Execution:** Run strategies with real signals and verify trade placement
- 3. **Position Synchronization:** Confirm database positions match Pacifica account state
- 4. **Error Handling:** Add graceful degradation when WebSocket disconnects
- 5. **Performance Monitoring:** Add metrics for cache hit rates and response times

Testing Workflow



Next Steps Priority

- 1. **High Priority:** Verify WebSocket price integration in trading decisions
- 2. **High Priority:** Test end-to-end trade execution flow
- 3. **Medium Priority:** Add comprehensive error handling and logging
- 4. **Medium Priority:** Implement automated position synchronization
- 5. **Low Priority:** Add performance monitoring and alerting trading_bot_workflow_diagram.md